The Digi-Comp II: an illustrated manual

West bay 3: how the Digi-Comp II computes, operation by operation · 1 November 2026

How Babbage's Hall's Digi-Comp II computes, operation by operation: count, clear, add, multiply, complement, subtract and divide, and what follows from them, and running by itself (AUTO), with the parts each one needs, for anyone who wants to build only some of it.

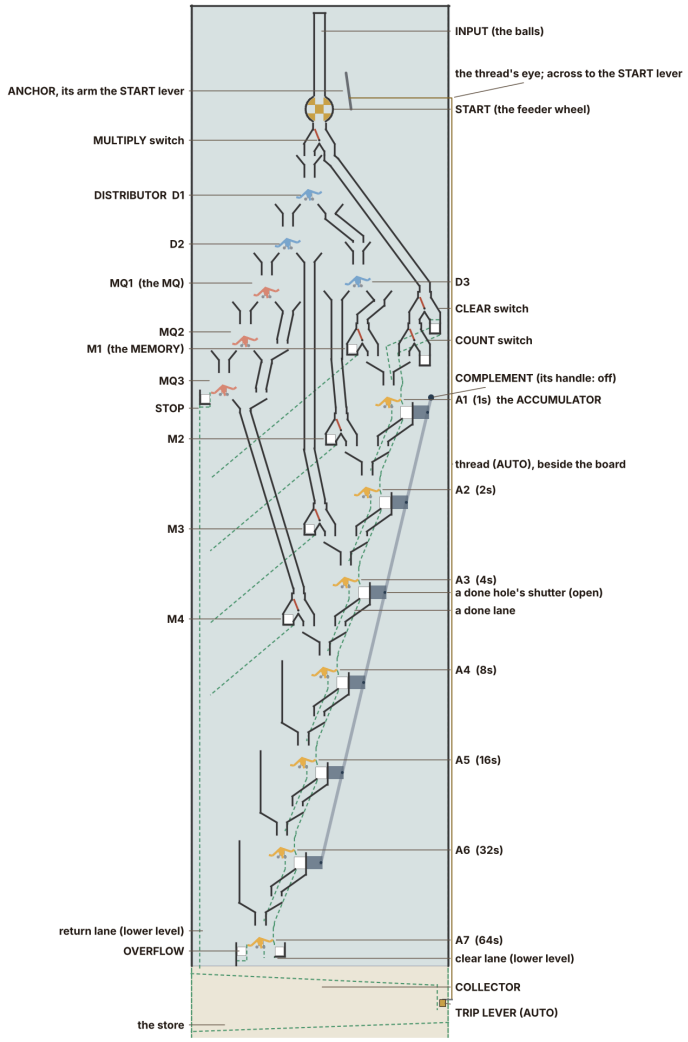
Every picture of the board is drawn from `digicomp2-geometry.lua`, the same numbers the simulation and the printed parts use (`manual/make-figures.sh` redraws them); only the arrows and labels are added. Where this board does a thing differently from E.S.R.'s, the original's way is given too, from its manual (Evil Mad Scientist's copy: cdn.evilmadscientist.com/KitInstrux/DCII-manual.pdf).

The whole idea in one paragraph. A flip-flop is a rocker that a ball always tips over, and the ball leaves on the side the rocker used to read. Chain them so that a ball leaving one side goes on to the next, and a ball adds one in binary, carrying as it goes: that is the accumulator. Let a set of switches (the memory) choose where balls go into it, and they add a number. Let three flip-flops (the distributor) deal the balls out to those switches in turn, and every four balls add the memory once. Let three more (the MQ) count those turns down, and stop when they run out, and the machine multiplies. Cover the holes where balls finish, so a ball goes on through every flip-flop and turns each one over, and the machine finds a number's complement, which subtracts. Everything else on the board is lanes and holes.

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1. The parts

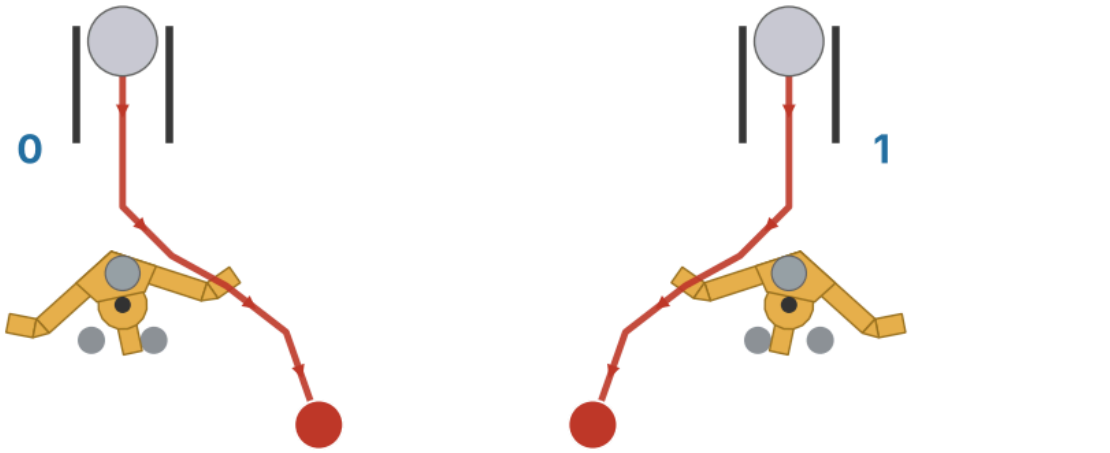


The board, seen from above. The upper plate (blue-grey) carries the balls while they compute; the lower level (dashed green walls) carries finished balls to the collector.

The board slopes toward you at 15 degrees, so a ball let go at the top rolls down it. It has two levels. The balls compute on the **upper plate**. A ball that has finished its work drops through a hole to the **lower level** and rolls under everything else to the **collector** at the bottom. Every ball ends there.

There are only two kinds of moving part, the flip-flop and the switch.

The flip-flop



A flip-flop: found at 0, the ball leaves right and the rocker turns to 1; found at 1, it leaves left and the rocker turns to 0.

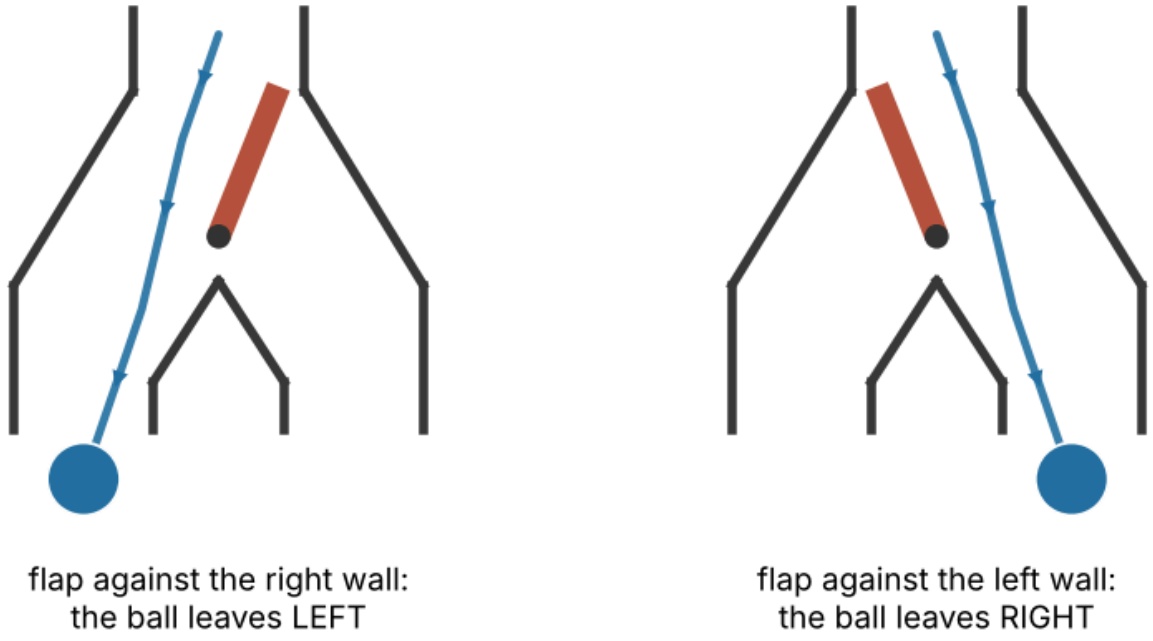
A **flip-flop** is a rocker on a pin: a roof with a cup at each end, a small steel ball pressed in under its apex for weight, and a tail that rests on one of two stop pegs. Its weight sits a little above the pin, so it falls over to one side or the other and stays there. That is one bit of memory: **apex right reads 1, apex left reads 0** (the 0 and 1 are printed on the plate on either side of each apex).

A ball rolling down onto it meets the roof on its raised side, rolls into that side's cup, and its weight tips the rocker over; at the end of the swing the cup tips the ball out. So **every ball flips the rocker**, and **leaves on the side the rocker used to read**:

The rocker read	It now reads	The ball leaves
0	1	to the right
1	0	to the left

That is all a flip-flop does. What it is used for depends on where its two exits lead. All thirteen on this board are the same printed part.

The switch



A switch: the flap closes one outlet; the ball takes the other.

A **switch** is a flap on a pin between two outlets, set by hand. Its free end rests against one wall of the inlet and closes the outlet on that side. A ball only pushes the flap harder against the wall and leaves by the other outlet, so the switch stays as it was set. Each switch has an "on" side, printed on the plate.

The registers and switches

Name	What it is	What it does
START	the feeder: a wheel with four notches under the **INPUT** channel of waiting balls, turned by a weight and held by an anchor	each push on the START lever (the anchor's arm) lets the wheel a quarter turn round: one ball out
AUTO/MANUAL	the thread from the trip lever at the bottom, hooked into the START lever or off	AUTO: each ball reaching the bottom lets the next go. MANUAL: one ball per push
MULTIPLY	a switch	on: the ball goes to the distributor. Off: down to CLEAR
CLEAR	a switch	on: the ball drops into the clear lane. Off: on to COUNT
COUNT	a switch	on: the ball goes into A1. Off: it drops out
A, the accumulator	seven flip-flops, A1 (1s) to A7 (64s)	holds the answer, 0 to 127
M, the memory	four switches, M1 (1s) to M4 (8s)	holds a number, 0 to 15, set by hand
MQ, multiplier-quotient	three flip-flops, MQ1 (1s) to MQ3 (4s)	counts down by one each cycle; 0 to 7
D, the distributor	three flip-flops, D1 to D3	deals the balls out four ways, in turn
STOP	a hole beside MQ3	a ball dropping through it means "finished"

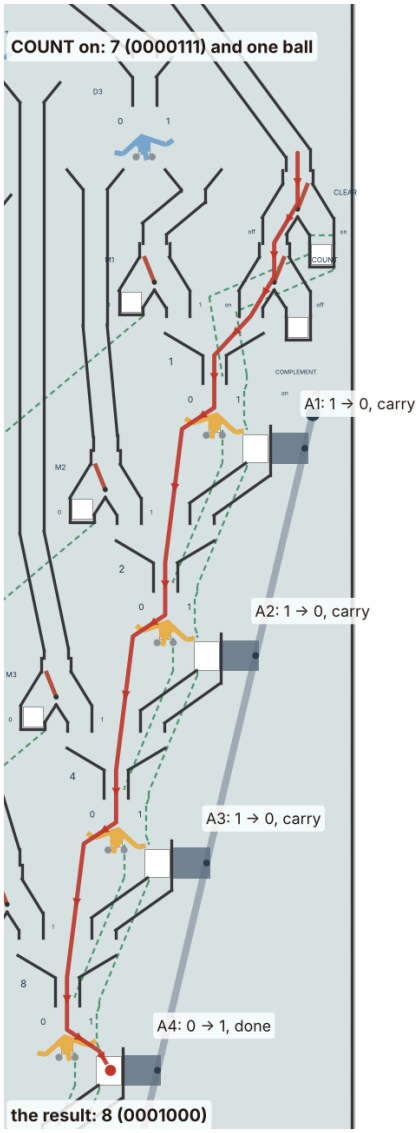
Name	What it is	What it does
OVERFLOW	a hole beside A7, with a shutter over it	a carry out of A7 (past 127) goes there. RUN (shutter closed): it rolls on, to the collector. HALT (open): it drops into a pocket, and in AUTO the machine stops
COMPLEMENT	six shutters over A1 to A6's done holes, on one bar with a handle	on: a ball goes on through every flip-flop below where it comes in, turning each over (section 6)

Reading binary

Each flip-flop of A stands for a power of two: A1 the 1s, A2 the 2s, A3 the 4s, and on to A7, the 64s (the numbers are printed beside them). A number is the sum of the places that read 1. Written the usual way, the 64s first: **0101101** is 32 + 8 + 4 + 1 = **45**. The sign beside the machine shows each register in binary and as a number, but the rockers are the machine's own reading.

2. Counting

Set up: A as you want it (0 to start), MULTIPLY off, CLEAR off, COUNT on. **Then:** START, once for each ball.



Counting: 7 (0000111) and one ball. A1, A2 and A3 each turn 1 to 0 and pass the ball on (the carry); A4 turns 0 to 1 and the ball drops out through A4's hole. The result is 8 (0001000).

The accumulator's flip-flops form a staircase. Each one's **right** exit is a hole to the lower level, so the ball is finished. Its **left** exit runs down into the next flip-flop. So a ball arriving at a flip-flop:

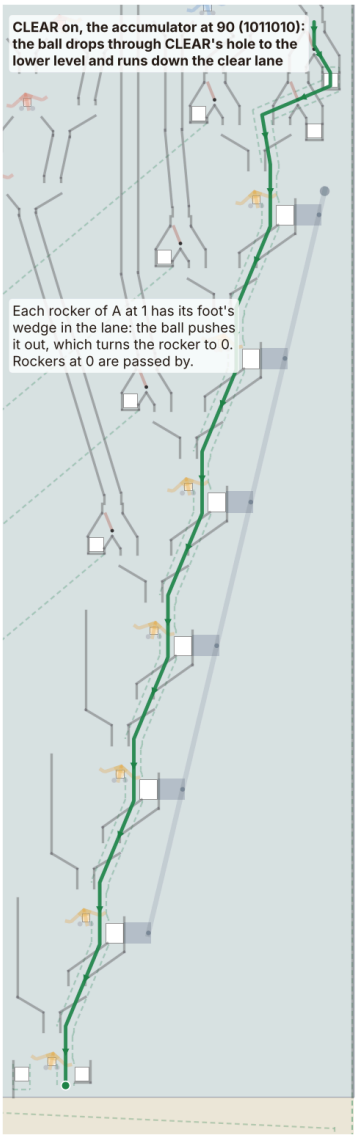
- finds it at **0**: sets it to 1 and drops out. One has been added.
- finds it at **1**: sets it to 0 and goes on to the next flip-flop, which is the **carry** (1 + 1 = 10 in binary).

A ball let into A1 therefore **adds one**, carrying as far as it must. A ball let into A3 adds four, and so on: the memory uses this when it adds.

Overflow. At 127 (1111111) the next ball carries through all seven and out of A7 through the OVERFLOW hole, leaving 0000000.

3. Clearing

Set up: MULTIPLY off, CLEAR on. **Then:** START, one ball.



CLEAR from 90: the ball drops through CLEAR's hole and runs down the clear lane on the lower level, past the foot of each of A's flip-flops.

Under each flip-flop of A, on the lower level, is its **foot**: an arm joined to the rocker through the plate, ending in a wedge. When the rocker reads 1, its wedge leans into the **clear lane**. A ball sent down the clear lane pushes every wedge it meets back out of its way, and pushing the wedge turns the rocker to 0. Rockers already at 0 are passed by. **One ball clears the whole accumulator.**

(You can also clear by hand: set every rocker of A to 0.)

4. Adding

Set up: A = the first number; M = the second (at most 15); MQ = 1; the distributor set to start (D1 = 1, D2 = 1, D3 = 0; key `D`); MULTIPLY on. **Then:** START until a ball drops through STOP: **five balls.**

Adding is a multiply by 1 (section 5): the first four balls add M to A, and the fifth finds the MQ used up and stops. Three things make it work.

The memory. Each memory switch stands beside one of A1 to A4. Set to 1, it sends a ball that reaches it into that flip-flop, which adds 1, 2, 4 or 8 there (with any carry). Set to 0, it drops the ball out. The

memory itself never changes.

The distributor deals the balls out in turn. D1 sends them to D2 and D3 alternately, and each of those alternately to its two sides: four balls, four places, then round again. Set to start, the four balls of each cycle go:

Ball of the cycle	Way through the distributor	Then
1st	D1 left, D2 left	through the MQ (one taken away), on to M4
2nd	D1 right, D3 right	M1
3rd	D1 left, D2 right	M3
4th	D1 right, D3 left	M2

So **every four balls add the memory to A once.**

The MQ counts the cycles (section 5).

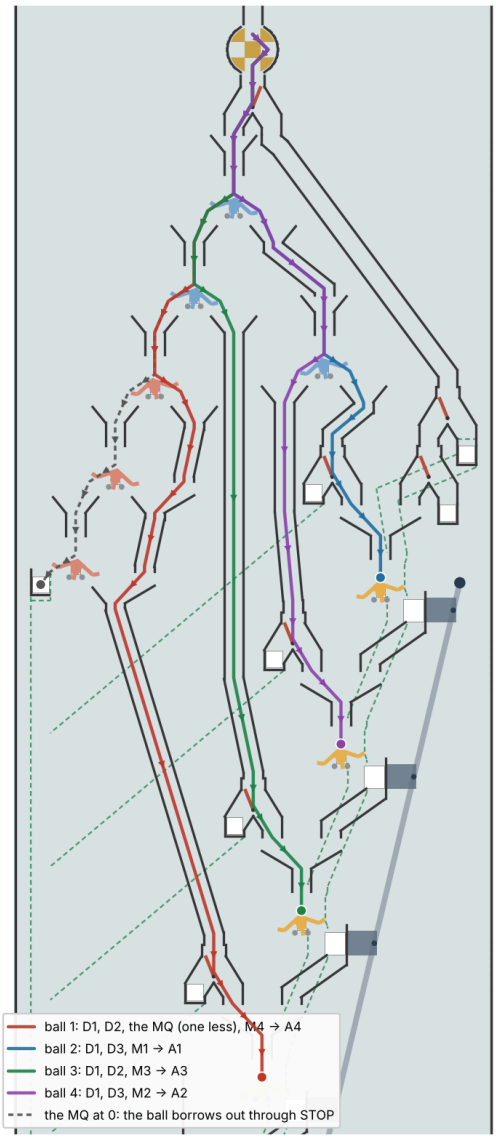
Worked example: 37 + 11 (the tour's). A = 0100101 (37), M = 1011 (11: M4, M2, M1 on; M3 off), MQ = 1:

Ball	Goes to	A after it	MQ after it
1	the MQ (1 to 0), then M4 = 1: adds 8	0101101 = 45	0
2	M1 = 1: adds 1	0101110 = 46	0
3	M3 = 0: drops out	0101110 = 46	0
4	M2 = 1: adds 2 (carries through A2, A3, A4 to A5)	0110000 = 48	0
5	the MQ, at 0: through STOP	**0110000 = 48**	7

Keep the sum at 127 or less (the total passes through OVERFLOW otherwise).

5. Multiplying

Set up: A = 0 (CLEAR it); M = the multiplicand (at most 15); MQ = the multiplier (at most 7); distributor set to start; MULTIPLY on. **Then:** START until a ball drops through STOP: **4 x MQ + 1 balls.**



One cycle of four balls with MULTIPLY on (D1 = 1, D2 = 1, D3 = 0 at the start), and, dashed, the ball that finds the MQ at 0 and leaves through STOP.

Multiplying is adding over and over: every four balls add M to A, and the MQ counts how many times.

The MQ is three flip-flops in a staircase like A's, but it is read the other way round: **1 with the apex to the left** (printed so on the plate). Read that way, a ball through it **takes one away**. A ball that finds an MQ flip-flop at 1 sets it to 0 and leaves to the right, which means done; it runs on to M4. One that finds it at 0 sets it to 1 and goes on to the next, which is the **borrow** (10 - 1 = 01). A borrow out of MQ3 means the MQ was already 0. That ball drops through **STOP**, and the hand (you) stops pushing START (in AUTO the machine stops by itself: the ball through STOP never reaches the trip lever).

Worked example: 13 x 3 (the tour's). M = 1101 (13: M4, M3, M1 on), MQ = 3, A = 0:

Balls	Cycle	A after the cycle	MQ
1 to 4	first: adds 8, 1, 4, 0	13	2
5 to 8	second	26	1
9 to 12	third	39	0
13	the MQ at 0: STOP	**0100111 = 39**	7

Put the smaller number in the MQ: 5 x 7 takes 21 balls with 5 in the MQ, 29 with 7.

What else multiplying does

These need nothing more than the multiply set-up, with different numbers.

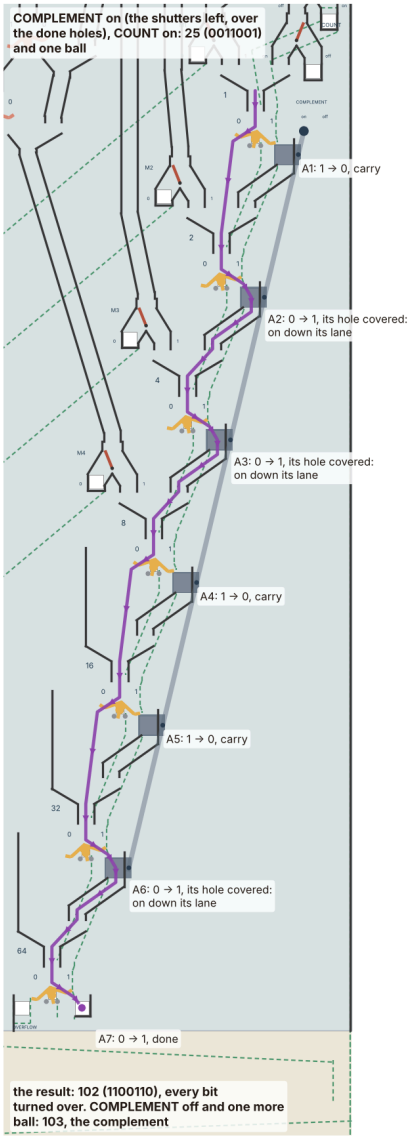
- **Multiply and add.** A isn't cleared by MULTIPLY, so whatever is in A is added to: A + M x MQ. Two multiplies in a row add up: 4 x 14, then 5 x 13 without clearing, gives 56 + 65 = **121**.
- **Squaring.** The same number in M and the MQ: 7 x 7 = **49** (29 balls).
- **Shifting.** Multiplying by 2, 4 or 8 moves a number's bits one, two or three places left: put 2, 4 or 8 in M and the number in the MQ, or the number in M and 2 or 4 in the MQ. M = 1101 (13), MQ = 4: **0110100** (52).
- **Decimal to binary** (from E.S.R.'s manual). To find 57 in binary, write it as 5 x 10 + 7: A = 7, M = 10 (1010), MQ = 5. Twenty-one balls later A reads **0111001**. (The MQ's 7 limits this to 79; split bigger numbers, as the original manual does for 123 = 64 + 59.)

6. The complement: subtracting, counting out, dividing

The complement

Subtracting is done by adding a **complement**. The two's complement of a 7-bit number is what you add to it to make 128 (0000000 with an overflow). Turn every bit over (the **ones' complement**), then add one:

	binary	
25	0011001	
every bit turned over	1100110	the ones' complement: 102, which is 127 - 25
plus one	**1100111**	the complement: 103, which is 128 - 25



COMPLEMENT on: one ball goes through all seven flip-flops of A and turns each one over. Where it sets one to 1, the done hole is covered, so it rolls on down that one's done lane into the next.

On this board. Each of A1 to A6 has, below its done hole, a **done lane** that runs down and across into the next flip-flop's funnel. The hole is open as a rule, and a ball that sets the flip-flop to 1 drops through it, done; the lane below is never reached. The **COMPLEMENT** switch is six **shutters**, one over each of those holes, all on one bar (its handle is beside A1): slid to the left (on), they cover the holes. Now a ball that sets a flip-flop to 1 rolls over the covered hole and down the done lane into the next flip-flop, just as one that sets it to 0 goes on to the next by the carry. Either way it goes on, so it turns over every flip-flop from where it came in to A7. (Out of A7 it drops through A7's done hole, which has no shutter, or through OVERFLOW.)

1. A = the number (count it in, or set it). COUNT on, MULTIPLY and CLEAR off.
2. COMPLEMENT on, START: **one ball**. A now holds 127 - A.
3. COMPLEMENT off, START: **one ball** adds one. A holds 128 - A, the complement.

On the original it is one START with COMPLEMENT on, and the machine does four balls by itself (in AUTO): six **mode** flip-flops T1 to T6 between A's flip-flops, and two more, CF1 and CF2, to sequence them. Ball 1 sets T1 to T6 to COMPLEMENT mode, ball 2 passes every bit of A and turns each over, ball 3

sets them back to ADD mode, ball 4 adds one. This board's AUTO has no CF1 and CF2, so the hand does what they do: it turns COMPLEMENT off between the two balls, in MANUAL. The shutters do the work of T1 to T6.

Negative numbers. Read this way, a number whose leftmost bit is 1 is negative: 1100111 is -25. Seven bits hold -64 to +63.

Subtracting

A = the number to take away. Complement it (two balls, as above). Then MULTIPLY on, M = the number to take it from, MQ = 1, D set to start, START until STOP (five balls: an add). With COMPLEMENT off the adding is ordinary.

Example: 12 - 4 (in the tour). A = 4 (0000100); complemented: 1111100 (124). Add M = 12: 124 + 12 = 136, which goes over 128: the ball that carries out of A7 drops through OVERFLOW, and A is left with **0001000 = 8**. The overflow is simply lost, which is what makes the complement subtract.

Counting out a number of balls

A = N; complement it (two balls); then COUNT on, MULTIPLY off. A counts up from 128 - N, and the **N-th ball** carries out of A7 through OVERFLOW. Stop there: that is N balls. In MANUAL the hand watches for that ball (`programs/count-out-9.dc2`). In AUTO, E.S.R.'s way, set OVERFLOW to HALT and press START once: the N-th ball drops into OVERFLOW's pocket, reaches no trip lever, and the machine stops itself (`programs/count-out-auto.dc2`).

Dividing

Dividing is subtracting over and over, counting how many times: adding the divisor to the dividend's complement until the sum goes over 128, while the MQ counts the cycles.

1. A = the dividend; complement it (two balls).
2. M = the divisor, MQ = 7 (111), D set to start, MULTIPLY on.
3. START, ball after ball. Each cycle of four balls adds M to A and takes one from the MQ (its first ball). When a ball drops through **OVERFLOW**, go on to **the end of that cycle**: stop when the next ball would go to the MQ (D1 and D2 both read 1, as D sets them to start).
4. Read the MQ's **lower numbers** (the other reading of its rockers: lower = 7 - upper; upper 100 is lower 011): that many cycles were begun. - **A reads 0**: the divisor went exactly. The quotient is the cycles, the remainder 0. - **Otherwise** the last cycle went over: the quotient is one less than the cycles, and the **remainder is M - A**.

Example: 13 / 5 (in the tour). 13 complemented is 115. The cycles add 5: 120, 125, and in the third 130, which goes over (the ball that adds 4 drops through OVERFLOW) and leaves 2. The cycle ends; the MQ reads upper 100, lower 011 = 3 cycles. A is 2, not 0: quotient 3 - 1 = **2**, remainder 5 - 2 = **3**. (13 = 2 x 5 + 3.)

Example: 25 / 5. 25 complemented is 103; 108, 113, 118, 123, and in the fifth cycle 128: over, leaving 0. Five cycles, A = 0: quotient **5**, remainder 0.

Why to the end of the cycle. A cycle adds M one bit at a time (on this board its 8s, then 1s, 4s, 2s), and A may go over 128 on any of them. Stopping at that ball would leave the rest of M unadded, and the remainder wrong; finished, the cycle has added all of M, and A = cycles x M - dividend. (E.S.R.'s manual stops at the overflow, in AUTO, and has a chapter on remainders that could not be read here; this rule is this board's own, and it is what the tour checks.)

In AUTO, E.S.R.'s way: set it up as above, OVERFLOW to HALT, AUTO, and START once. The machine stops itself at the overflow, the overflowing ball in OVERFLOW's pocket, wherever in the cycle that was. If A reads 0 it went exactly, and the cycles (the MQ's lower numbers) are the quotient (`programs/divide-auto.dc2`: 25 / 5). Otherwise finish the cycle by hand, in MANUAL, and read it as above. (A halt can also leave A at 0 when it did not go exactly: 11 / 5 overflows on the first ball of its third cycle, the 1s, with A at 0, its 4s not yet added. So when in doubt, finish the cycle.)

Limits. With the MQ at 7, at most seven cycles: quotients up to 6 (7 when exact). If the MQ runs out first, a ball drops through STOP: the quotient was too big for it.

Running by itself: AUTO

On the original, AUTO lets a ball reaching the bottom roll under the START lever and release the next ball, so a whole multiply runs by itself. The ball through STOP takes a way that misses the lever, so the machine stops.

On this board a wound weight turns the feeder wheel, and an anchor holds it, as a clock's weight turns its wheels and its escapement holds them: each push on the anchor's long arm (the START lever) lets the wheel round a quarter turn, one ball. The weight is a cup of eight steel balls on a cord from a drum on the wheel's axle, under the board; a rotary damper on the axle's end (the fly) keeps the wheel to about 3 radians a second. The cup falls 6.3 mm a ball, 57 balls from wound up to the table: wind it by turning the drum's knurled flange backward (a ratchet lets it turn back alone). Wind it before a long AUTO run: 25 balls left is enough for any in the library. At the bottom every ball goes down the **trip channel** at the right, under the **trip lever**, and on to the store. A ball lifts the lever as it passes; an arm hanging from the lever's axle, beside the board, swings down the slope, and a **thread** tied to it, up beside the board and across, pulls the START lever as the hand would. In AUTO the thread is hooked into the START lever, and the anchor lets the next ball go; in MANUAL it is unhooked. The ball through STOP drops into a pocket under the plate, never reaches the lever, and the machine stops.

	AUTO	MANUAL
the thread	hooked into the START lever	unhooked, on its peg
START (`Space`)	once: the machine runs until a ball drops through STOP	once for each ball
the trip lever	trips the anchor: the next ball goes	lifts, and nothing follows

So a multiply in AUTO is: set it up as in section 5, AUTO (`U`), and one `Space`. **OVERFLOW** set to HALT (`0`) stops it too, at a carry out of A7: counting out a number of balls, and dividing, also run by themselves (section 6).

7. Which parts each operation needs



Which parts of this board each kind of operation uses.

Operation	A	CLEAR	COUNT	MULTIPLY	MEMORY	COMPLEMENT	OVERFLOW	AUTO
Count	yes		yes					
Clear	yes	yes						
Add	yes			yes	yes			
Multiply, multiply-add, square, shift	yes			yes	yes			
Decimal to binary	yes			yes	yes			
Complement	yes		yes			yes		
Subtract	yes		yes	yes	yes	yes		
Count out N balls	yes		yes			yes	to halt	
Divide	yes		yes	yes	yes	yes	to halt	
Any of them running by itself								yes

The column names are the parts a program names on its `PARTS` line (section 9), and what each takes in:

- **A**: the seven flip-flops, their channels, funnels and stop pegs, the guides, the overflow hole and its pocket, and two things that look like other parts' but do A's work too: the clear lane on the lower level (its walls bring the balls that drop out of A's done holes down to the collector moving fast enough to roll on into the trip channel), and the wall beside each long done hole, the first stretch of its done lane (a fast carry would skip across the hole without it); the overflow hole's outer wall likewise.
- **CLEAR**: the CLEAR switch and its drop into the clear lane.
- **COUNT**: the COUNT switch and its drop.
- **MULTIPLY**: the distributor (D1 .. D3), the MQ, their channels, the lanes from them to the memory, STOP and its pocket.
- **MEMORY**: the four memory switches, their drop holes' walls and the walls under them that take what they drop to the return lane.
- **COMPLEMENT**: the six shutters, their bar and the done lanes, but for their first stretch.
- **OVERFLOW**: its shutter and the lane over it (RUN), past the hole. Only needed to halt the machine in AUTO: without the shutter the overflow hole is always open, the carry out of A7 drops into its pocket (A's), and the hand sees it there as well as anywhere (`UNTIL OVERFLOW`).
- **AUTO**: the trip lever, its hanging arm, the eye and the thread.

Always there: the board, its rim and the collector, and the feeder (the wheel, the anchor, the weight and the damper) with the MULTIPLY switch, which every ball passes on its way down (off, it sends the ball on to CLEAR). The CLEAR switch and the lane to it are there for CLEAR or COUNT: a ball for COUNT passes CLEAR's switch too.

Building only part of it

In the exhibit, each program in the library runs with only the parts its `PARTS` line names on the board: the others are taken off (out of the simulation, not only out of sight), as they would be on a machine built only that far. `R` by hand puts every part back. (The plate's holes and its lettering stay: they are cut in the plate.)

The steps this machine was built in are each a complete, smaller machine, kept in the repository's history. Each has its own board, tiles, parts list and `make-stls.sh`:

To build	Take	It does	Board
A three-bit counter, one level, balls into bins	step 1 (commit `cb8bb07`)	count to 7	the smallest
The accumulator with START, CLEAR and COUNT, two levels	step 2a (commit `201ee23`)	count to 127, clear	322 × 878 mm, 10 + 10 tiles
Count, clear, add, multiply	step 2b (commit `71d155a`)	and section 5's uses	424 × 1216 mm, 14 + 14 tiles
Count, clear, add, multiply, complement, subtract, divide	step 2c (commit `1c191be`)	all of the above, by hand	424 × 1621 mm, 18 + 18 tiles
Count, clear, add, multiply, complement, subtract, divide; AUTO (the weight a torque)	step 2d (commit `44b5a1b`)	and AUTO, stopped by STOP	424 × 1691 mm, 18 + 18 tiles
Everything so far	this folder (step 2f)	and OVERFLOW's halt; the weight and the fly as parts	424 × 1691 mm, 18 + 18 tiles

To get an earlier step's files without changing your checkout:

```
git archive 201ee23 demo/WyomingWill/BabbagesHall/exhibits/DigiComp2 | tar -x -C /some/other/folder
```

Its README lists the parts to print and buy. Some other ways to build less:

- **Without CLEAR**, clear by hand (set every rocker to 0). The feet are part of every rocker either way; the clear lane's walls would have to be taken out of `digicomp2-geometry.lua` to leave them off the lower tiles.
- **Fewer memory bits or a shorter MQ** limit the numbers (M up to 15, MQ up to 7 now). That would need the layout redrawn: it is set in `digicomp2-geometry.lua` and has only been built and tested as it is.
- **For adding only**, the MQ is still needed: it is what stops the fifth ball.
- **Without COMPLEMENT**, leave out the shutters and their bar: the done holes stay open, the done lanes are never reached, and the board is what step 2b's was, only longer. (A shorter board without the lanes is step 2b's.)

8. Settings at a glance

Operation	A	M	MQ	D	MULTIPLY	CLEAR	COUNT	Balls
Count	start value				off	off	on	one each
Clear					off	on		1
Add A + M	A	M	1	start	on			5
Multiply M x MQ	0	M	MQ	start	on			4 x MQ + 1
Multiply-add A + M x MQ	A	M	MQ	start	on			4 x MQ + 1
Complement A	A				off	off	on	1 with COMPLEMENT on, 1 with it off
Subtract M - A	complement of A	M	1	start	on			5
Count out N	complement of N				off	off	on	until OVERFLOW: N (in AUTO, OVERFLOW HALT)
Divide A / M	complement of A	M	7	start	on			until OVERFLOW, then to the end of that cycle (in AUTO, OVERFLOW HALT)

On this board, by hand (keys in the simulation): A's rockers `1` to `7`, the memory `H J K L`, the MQ `8 9 0`, the distributor to start `D`, MULTIPLY `M`, CLEAR `C`, COUNT `N`, COMPLEMENT `X`, OVERFLOW `O`, AUTO/MANUAL `U`, START `Space`. The tour (`T`) runs the program library (section 9), one after another: counting, clearing, 37 + 11, 13 x 3, 13 x 5 in AUTO, the complement of 25, 12 - 4, 13 / 5, 25 / 5 by hand and in AUTO, and counting out nine by hand and in AUTO. Before each ball, in the tour or with `Space`, the exhibit draws the way the ball will go as a yellow dashed line on the board, and its sign says what the ball will do and what the accumulator was and will be: a good way to follow this manual on the machine itself.

9. Programs: the operator's part, written down

The machine has no stored program. Its registers (A, M, MQ) and the lanes between them are the arithmetic; its switches (MULTIPLY, CLEAR, COUNT, COMPLEMENT) choose which way a ball goes; each ball let in is one tick of its clock; STOP and OVERFLOW are its two signals to halt. What sets the switches between balls, and decides when to stop, is the operator (in AUTO, the machine stops itself at STOP). So an operation's procedure (sections 2 to 6) is like the microcode inside a computer's control unit: the settings and clock ticks that make one instruction, here kept in the manual and the operator's hands. The machine runs one loop by itself: with MULTIPLY on, the distributor deals the balls four ways and the MQ counts the cycles down to STOP.

On this board those procedures can be written down as **programs**, text files in `programs/`, and run by the exhibit's hand, line by line.

The language

Each line holds steps, separated by spaces; `;` begins a comment, to the end of the line. The first comment line is the program's title. The steps are the keys you would press, and a few more:

Step	What the hand does
`R`	reset: A, M and MQ 0, D set to start, COUNT on, MULTIPLY, CLEAR and COMPLEMENT off, OVERFLOW RUN, MANUAL
`1` .. `7`	turns A1 (1s) .. A7 (64s) over
`H` `J` `K` `L`	turns M1 .. M4 over
`8` `9` `0`	turns MQ1 .. MQ3 over
`D`	sets the distributor to begin a cycle
`M` `C` `N` `X`	MULTIPLY, CLEAR, COUNT, COMPLEMENT on ↔ off
`Space`	START: one ball (in MANUAL; in AUTO it is an error: use `RUN`)
`AUTO`, `MANUAL`	hooks the thread into the START lever, or takes it off
`RUN`	(in AUTO) START once, then waits while the machine runs, checking each ball as it lifts the trip lever, until one drops through STOP
`SET A n`, `SET M n`, `SET MQ n`	sets that register to n
`UNTIL STOP`	balls until one drops through STOP
`UNTIL OVERFLOW`	balls until one drops through OVERFLOW
`OVERFLOW HALT`, `OVERFLOW RUN`	opens OVERFLOW's shutter (a carry out of A7 drops into its pocket: in AUTO the machine stops), or closes it
`UNTIL CYCLE`	balls until the next would begin a cycle (D1 and D2 read 1); none if it would already
`CHECK A n`, `CHECK M n`, `CHECK MQ n`	says whether that register reads n
`PARTS name ...`	the parts the program needs (section 7: `A`, `CLEAR`, `COUNT`, `MULTIPLY`, `MEMORY`, `COMPLEMENT`, `OVERFLOW`, `AUTO`). While it runs the others are off the board, and a step that needs one it doesn't name is an error when the program is read. Without a `PARTS` line, the whole machine

Here is section 6's division, every ball written out (`programs/divide-13-by-5.dc2`):

```
; Divide: 13 / 5, ball by ball
; Every step written out, as the hand does it. The answer: the MQ's other
; reading (7 - MQ) is the cycles begun, 3; A is 2, not 0, so the quotient
; is 3 - 1 = 2 and the remainder M - A = 5 - 2 = 3.

; ---- set up ----
PARTS A COUNT COMPLEMENT MULTIPLY MEMORY   ; only these on the board
R            ; reset: A = 0, MQ = 0, M = 0, COUNT on, everything else off
1  3  4      ; A = 13 (1 + 4 + 8)                              A = 0001101 = 13

; ---- complement A (the hand does CF1 and CF2's job) ----
X            ; COMPLEMENT on: the done holes covered
Space        ; one ball through all seven bits, each turned over   A = 1110010 = 114
X            ; COMPLEMENT off
Space        ; one ball adds 1: 128 - 13                           A = 1110011 = 115
```

```
; ---- the divisor, the counter, the loop ----
H  K         ; M = 5 (its 1s and 4s)                              M = 0101 = 5
8  9  0      ; MQ = 7                                             MQ = 111 = 7
D            ; distributor set to begin a cycle
M            ; MULTIPLY on

; ---- run: each cycle of four balls takes 1 from the MQ and adds M to A ----
Space        ; cycle 1: MQ -> 6, M4 is 0                          A = 115
Space        ;          M1: +1                                    A = 116
Space        ;          M3: +4                                    A = 120
Space        ;          M2 is 0                                   A = 120
Space        ; cycle 2: MQ -> 5                                   A = 120
Space        ;          +1                                        A = 121
Space        ;          +4                                        A = 125
Space        ;          M2 is 0                                   A = 125
Space        ; cycle 3: MQ -> 4                                   A = 125
Space        ;          +1                                        A = 126
Space        ;          +4: 130, over 128: OVERFLOW               A = 2
Space        ;          M2 is 0: the end of the cycle             A = 2
             ; HALT: the next ball would start a cycle (D1 and D2 both read 1)
CHECK A 2
CHECK MQ 4
```

`programs/divide.dc2` does the same with loops (`UNTIL OVERFLOW`, then `UNTIL CYCLE`), for any dividend and divisor set in its two `SET` lines.

Running one

In the exhibit, the **program** slider chooses one from the library and `G` runs it (`G` again stops it). The tour (`T`, and as the exhibit starts) runs them all, one after another. While it runs, the program is on the **program board** behind the machine, the line being done lit (`B` brings the camera close enough to read it, and back); the listing is also in the Shortcuts pane, the line being done first and marked `->`, and the status bar at the bottom of the window shows that line. Every ball is shown before it goes (the yellow dashed line, the sign's THIS BALL) and checked after, as in the tour. To write your own, add a file to `programs/` and its name to `programs/LIBRARY` (the slider gives them in that order), and open the exhibit again.

The library

#	File	What it does
1	`count.dc2`	counts to 5
2	`clear.dc2`	clears A from 90
3	`add-37-11.dc2`	37 + 11
4	`multiply-13-by-3.dc2`	13 × 3
5	`multiply-auto.dc2`	13 × 5 in AUTO: one START, 21 balls by themselves
6	`complement-25.dc2`	the complement of 25
7	`subtract-12-4.dc2`	12 - 4
8	`divide-13-by-5.dc2`	13 / 5, ball by ball (the listing above)

#	File	What it does
9	`divide.dc2`	25 / 5, with loops
10	`divide-auto.dc2`	25 / 5 in AUTO, stopped by OVERFLOW (HALT)
11	`count-out-9.dc2`	counts out nine balls
12	`count-out-auto.dc2`	counts out nine balls in AUTO, stopped by OVERFLOW (HALT)